

Applied Data Science Project

L25 – Human Centred Design. Introduction: principles and tools.



**Politecnico
di Torino**

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AI, Data & Space | LINKS Foundation

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I'm interested in **the human dimension of technology innovation**. I've been working on **research and innovation** projects, leading activities related to the psychological and social aspects that **influence the adoption and interaction with new technologies**.

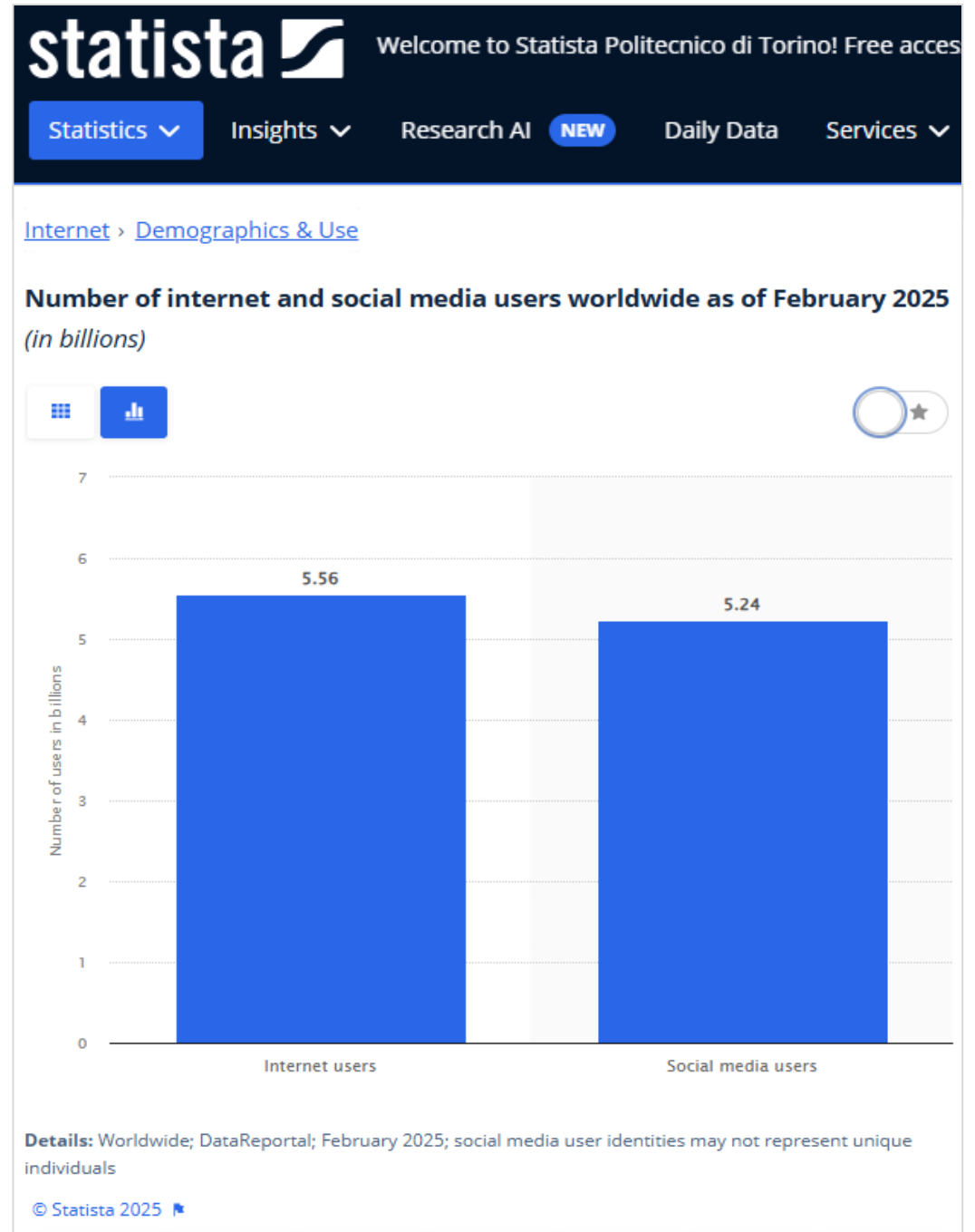
I design and assess innovative and people-centred services, applying the Human-Centred Design approach.

Why design is interesting for data science and viceversa?

Digital Population

As of February 2025, **5.56 billion** individuals **worldwide** were internet users, which amounted to **67.9% of the global population**.

Of this total, 5.24 billion, or 63.9% of the world's population, were social media users.



Why design is interesting for data science and viceversa?

We all daily produce tons of data,
even unintentionally
(preprint Shahriari & Ragan, 2025).

A SYSTEMATIC SURVEY OF EMPIRICAL USER STUDIES OF UNINTENTIONAL INFORMATION DISCLOSURE IN EVERYDAY DIGITAL INTERACTION

Reza Shahriari
University of Florida
Gainesville, FL
rshahriari@ufl.edu

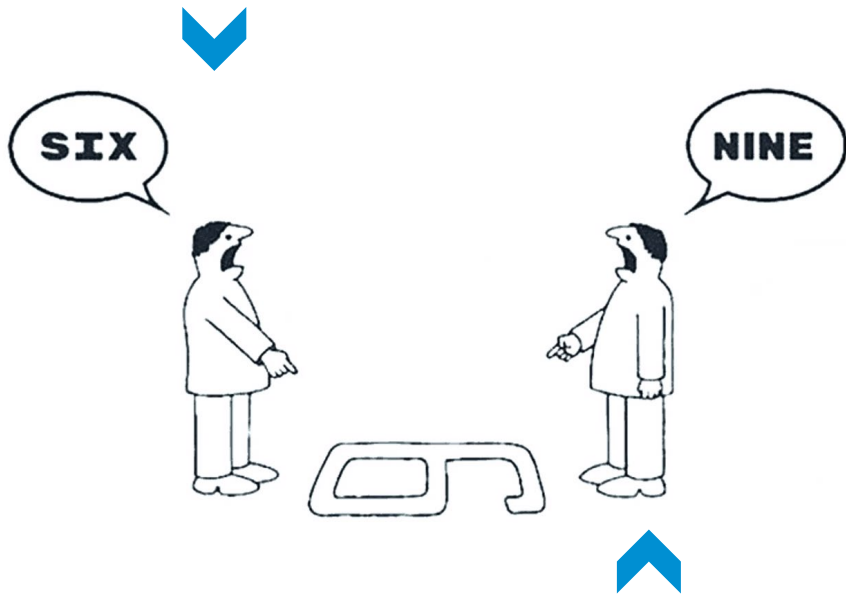
Eric D. Ragan
University of Florida
Gainesville, FL
eragan@ufl.edu

ABSTRACT

The exchange of personal information in digital environments poses significant risks, including identity theft, privacy breaches, and data misuse. Addressing these challenges requires a deep understanding of user behavior and mental models in diverse contexts. This paper presents a systematic literature review of empirical user studies on unintentional information disclosure in usable security, covering 101 papers published across six leading conferences from 2018 to 2023. The studies are categorized based on methodologies—quantitative and qualitative—and analyzed for their applications in various scenarios. Major subtopics, including data privacy, security in browsers, and privacy tools, are examined to highlight research trends and focal areas. This review provides details on topics and application areas that have received the most research attention. Moreover, by comparing descriptive and experimental approaches, findings aim to guide researchers of strategies to mitigate risks associated with online everyday interaction.

Why design is interesting for data science and viceversa?

Data science aims to **learn from data and extract value** from it, not only or necessarily economic value.



They share the **responsibility and power to**

- decide which data to collect and use
- provide data replying to real user needs
- collect and analyse data to test and improve services
- monitor services over time to improve them

Psychologists and Designers **turn data into experiences** that people can understand.

Widen our perspective.

IDEO

Data science is a discipline of human-centered design.

“When data science, interaction design, and engineering experts come together, we’re able to introduce radically new experiences and systems.”

DEAN MALMGREN
PARTNER, IDEO CHICAGO



Google Design

“Designers must be embedded in engineering and coding teams to keep the AI and machine learning efforts real—to keep them part of the world.”

Paola Antonelli
MoMA's Senior Design Curator



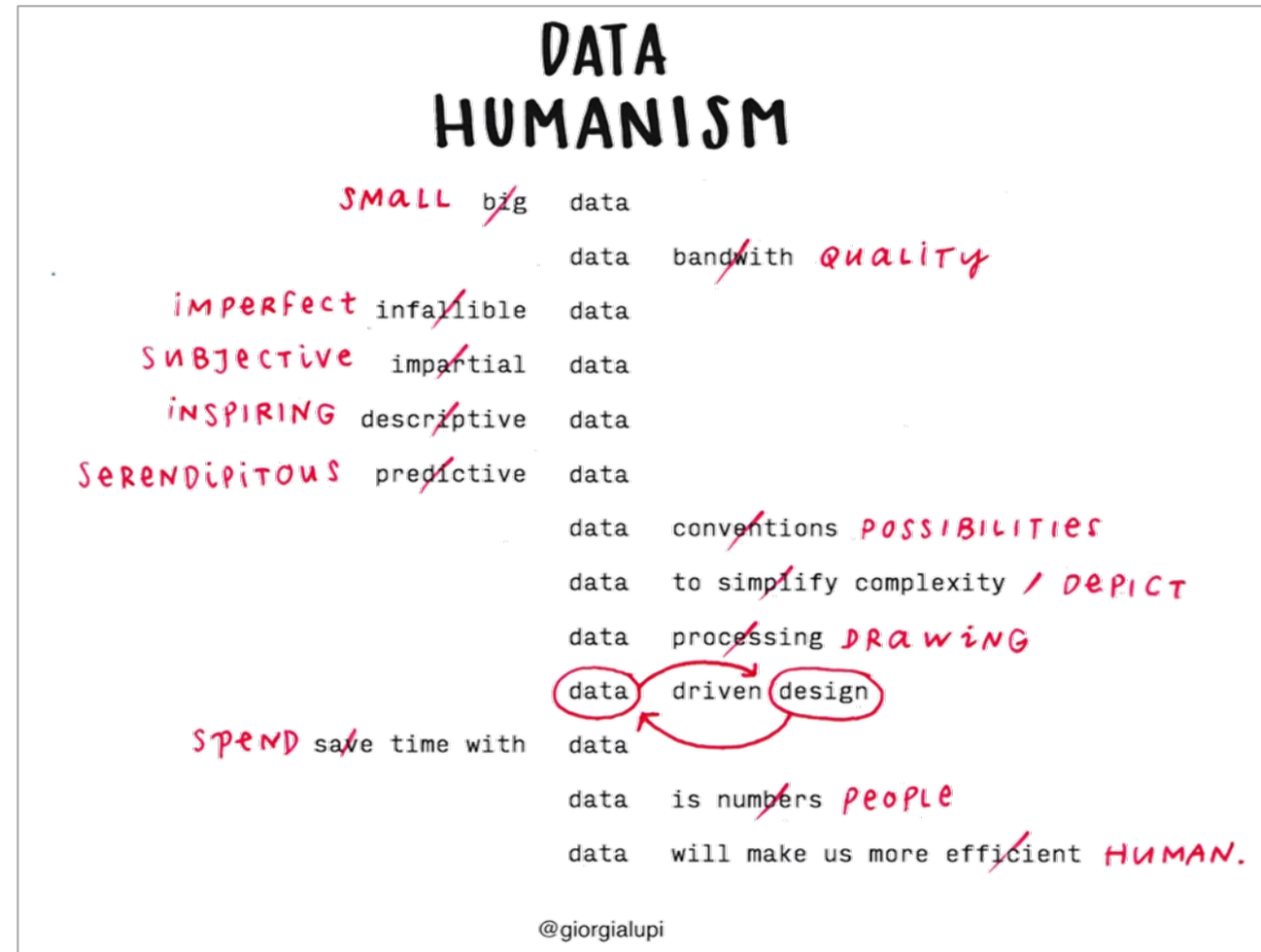
Not keeping into consideration the relationship between the digital tools we create/develop/manage and human behaviours, keeping on leaving those relations misunderstood and uncontrolled, might have **unintended consequences** and encourage the development of very negative phenomena for individuals, communities and populations.

Why this module?

To overcome «*the impersonality of a purely technical approach to data and to start designing ways to **connect data to what it really represents**: knowledge, behaviours, people.*»

Big Data is a commodity.

Data Humanisms offers a **new way to think about data, their usage, value, and limitations, informed by the real world, aware of human diversity, needs, and rights.**



Goals and Content of this module (6h)

- Get to know the **Human-Centred Design (HCD) principles** to nurture a wider and wiser project outlook
- **Apply some design tools** to frame data analysis around users' needs and desired outcomes
- Adopt a **design-oriented mindset** to the project.

29th of October 2025

**A humanistic approach to data
Hands-on: The stakeholders
analysis and mapping**

5th of November

**Modeling the audience
Hands-on: The users' Personas**

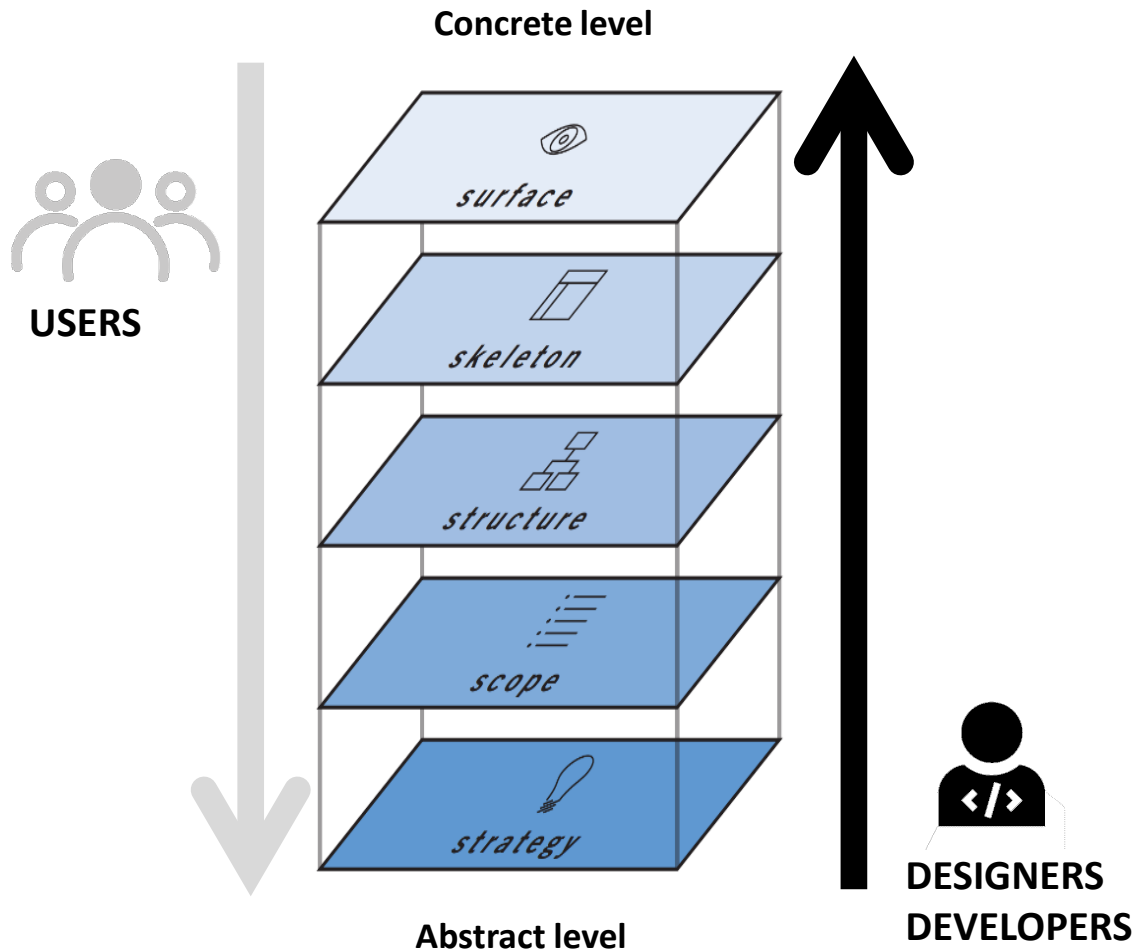
To apply to your projects

Data is experience

Experiences
are the events of **interaction with the world**
always involve a **touchpoint** (human, technological, symbolic, ...)
require the person's **presence** and/or participation
last **over time**
continue to have an **impact** even after they have happened.
They **change** us

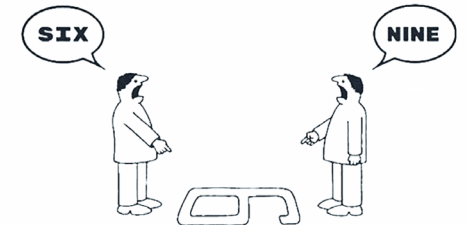


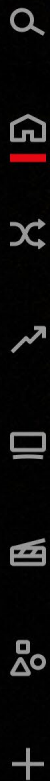
The experience is a process.



Experiences begin with the **surface**, where the person explores content and data within their **structure** to identify matches with their goals (**scopes**) and plans (**strategy**).

People and designers/technicians share the same path, but **in the opposite direction**.





N SERIES

STRANGER THINGS

TOP 10 #1 in TV Shows Today

When a young boy vanishes, a small town uncovers a mystery involving secret experiments, terrifying supernatural forces and one strange little girl.



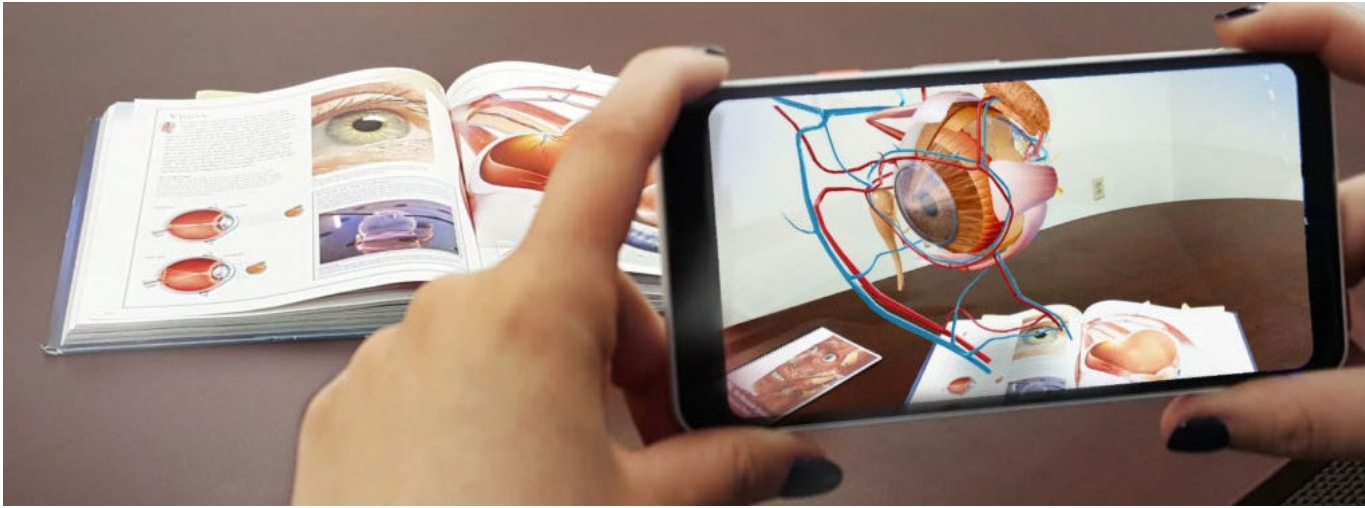
Play

More info

TV-14

Trending Now





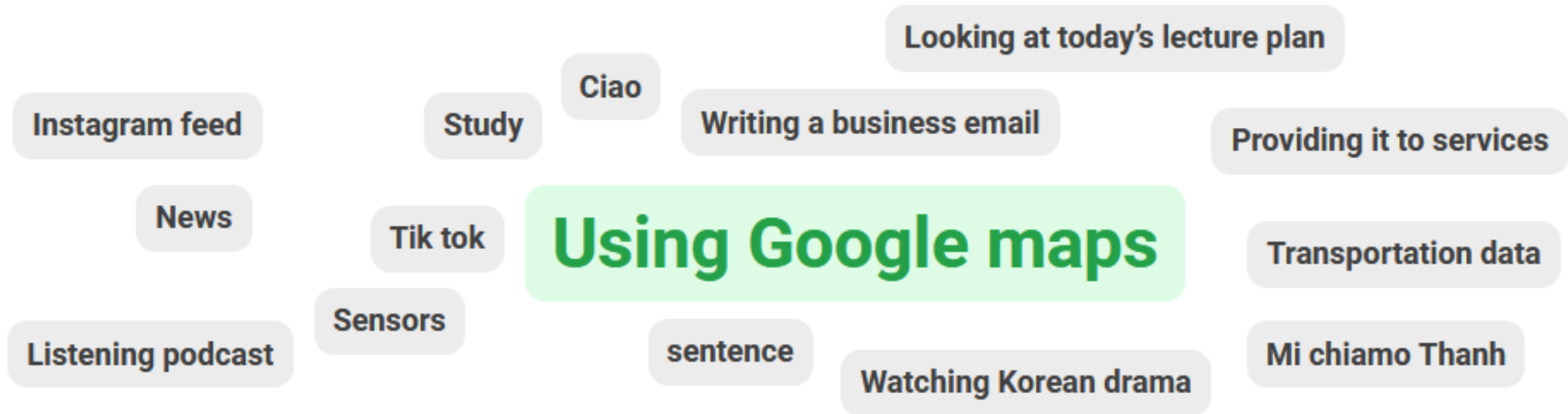
Data is **situated** within specific contexts and shapes the work practices and tools we use to accomplish various tasks (Bernhardt et al., 2022).



Bernhardt, A., Kresge, L., & Suleiman, R. (2022). **The Data-Driven Workplace and the Case for Worker Technology Rights.** *ILR Review*, 76(1), 3-29. <https://doi.org/10.1177/00197939221131558>

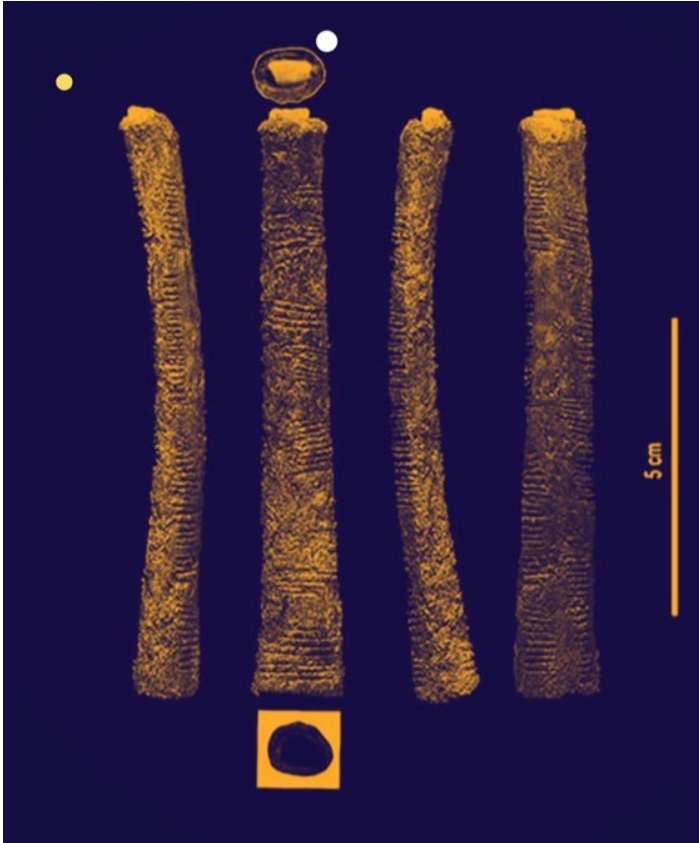
How do you experience data in your everyday life?

Review answers 17



20.000 A.C

Data has always been collected and observed.

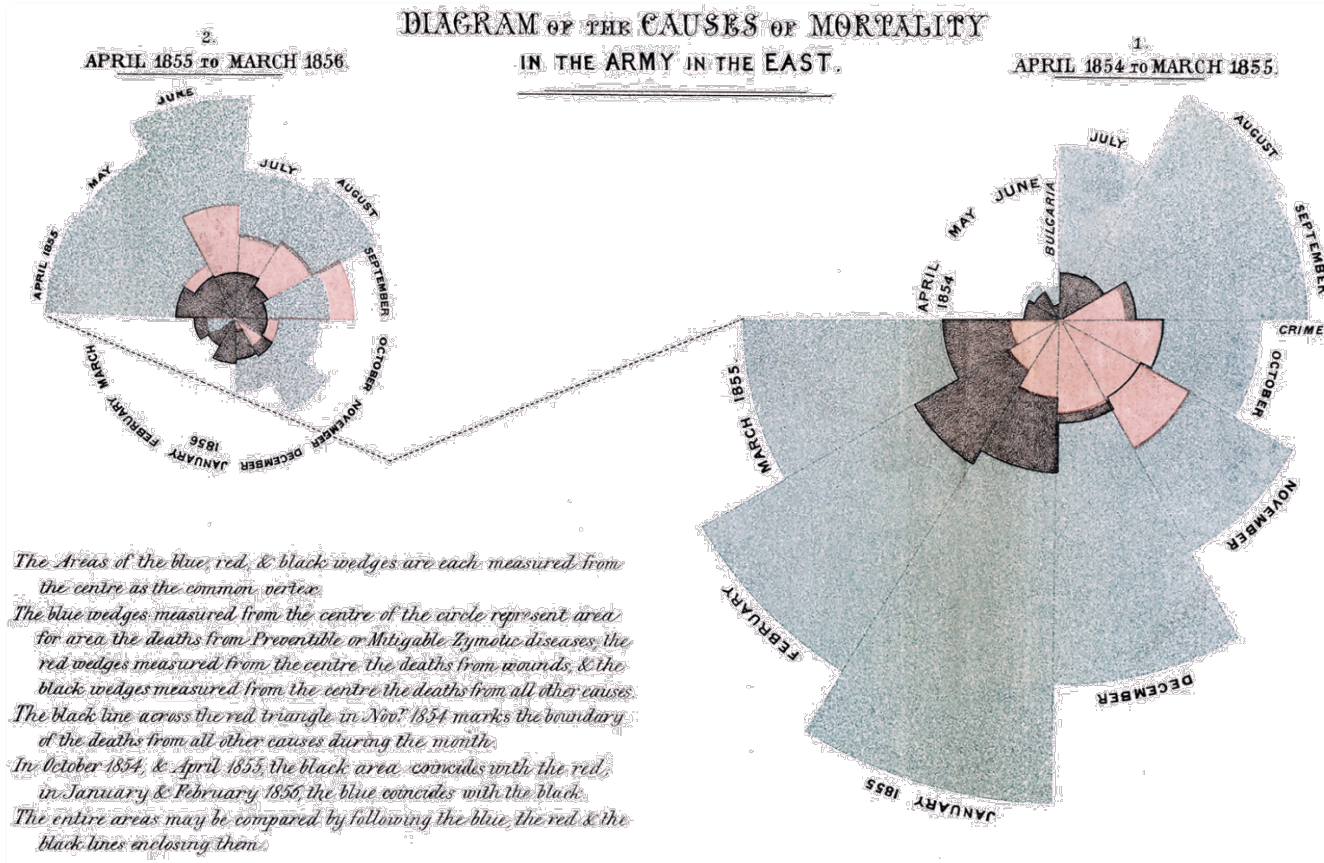


The first traces of data collection and calculation go back to the Palaeolithic (2.5 million – 12.000 years ago).



Baboon bones covered with signs grouped in 3 columns that occupy the entire length is an instrument **to track the lunar phases**, the ethno-mathematician Claudia Zaslavsky suggests.

1859

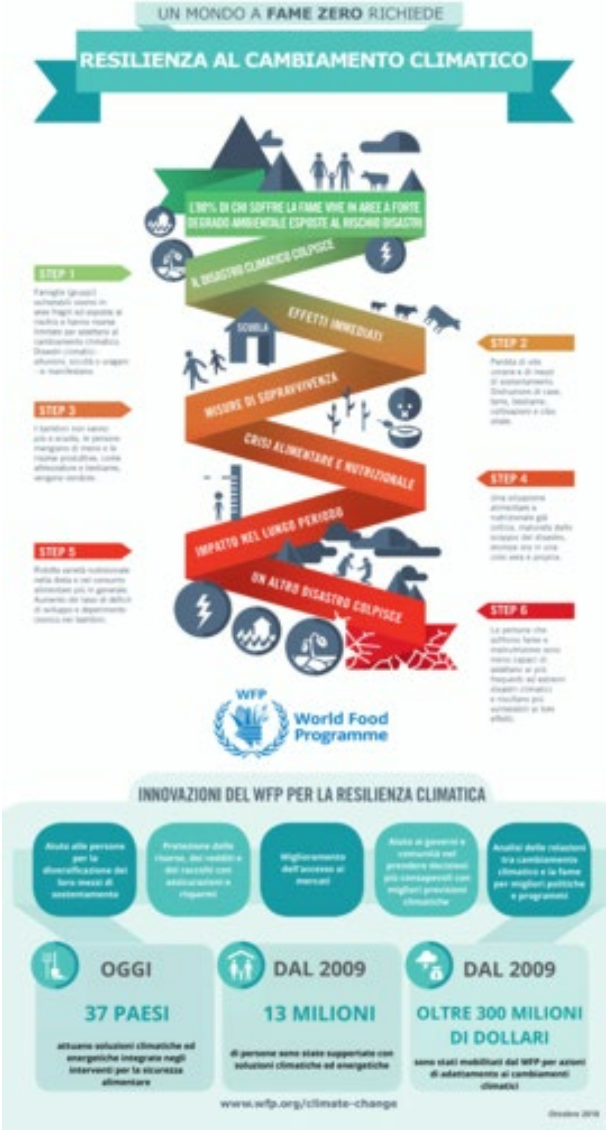
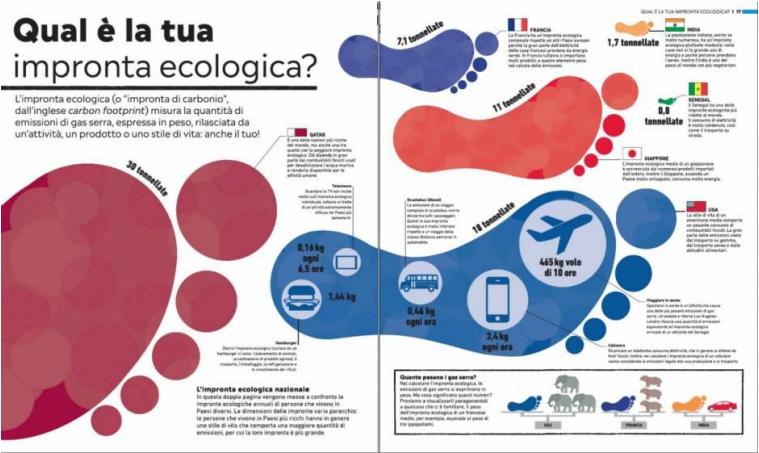
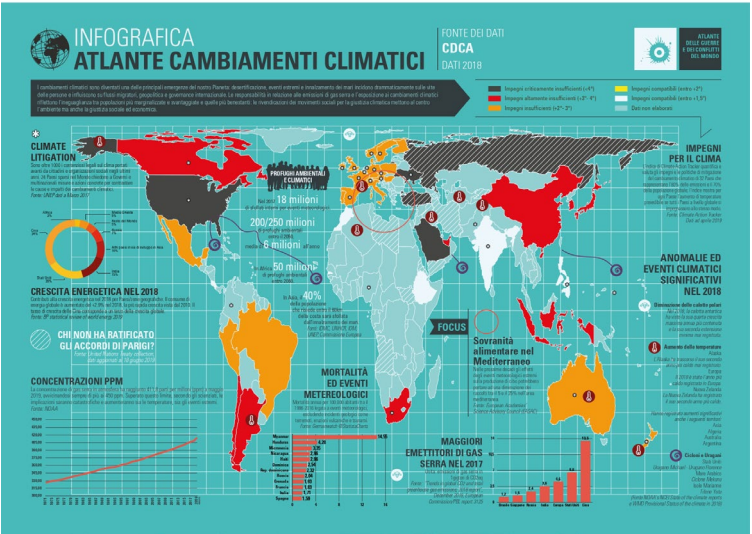
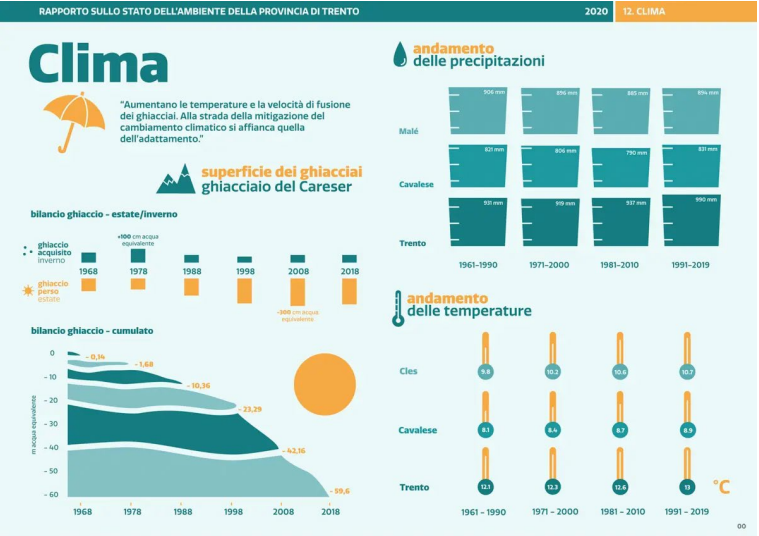


The first visualization of complex data.

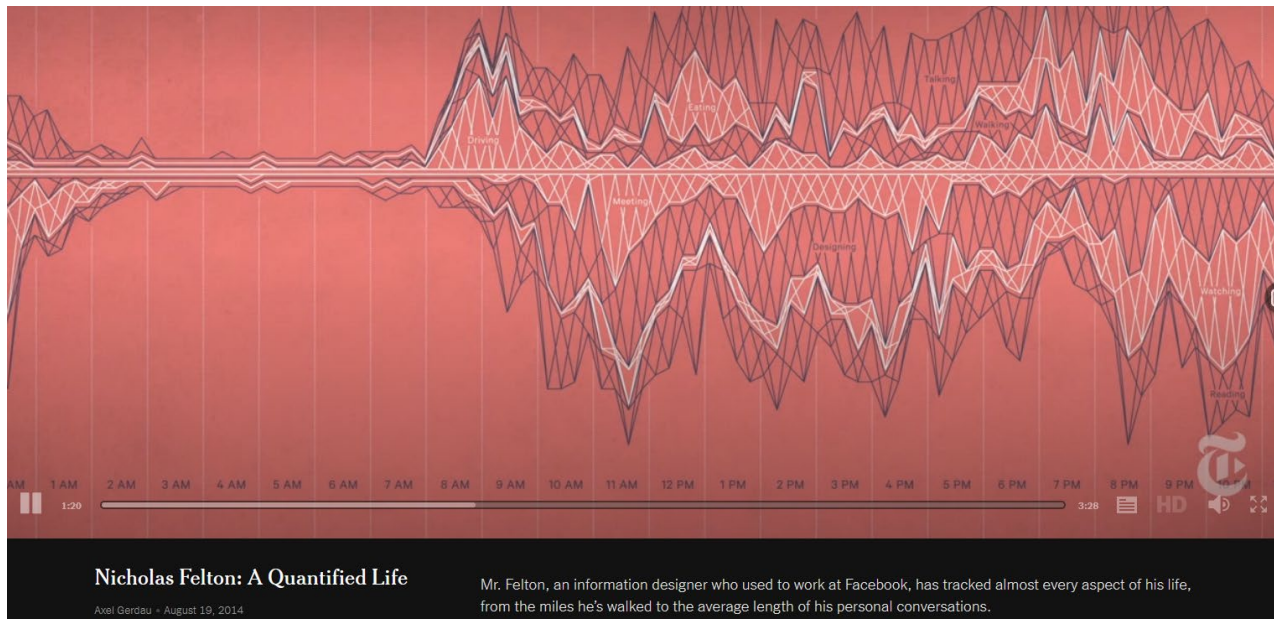
➤ The nurse Florence Nightingale began to track the causes of death of British soldiers in military hospitals during the Crimean war and **invented** the **polar chart** to display the pieces of evidence and communicate them.

1896...

The first work on global warming was published.



2014



Felton, 2014. The quantified self. The New York Times video, 2014.

The rapid adoption of **wearable tracking devices** and **motion-sensitive apps** has led to the development of the **“Quantified Self” movement (QS)**.

Scholars have begun to take notice and incorporate data from QS into the research and design of new learning environments (Lee, 2014).

Activities quantified:

- Diet
- Sport
- Books read
- Finance
- Dreams
- ...

Lee, V. R., **What's Happening in the "Quantified Self" Movement?** (2014). *Instructional Technology and Learning Sciences Faculty*

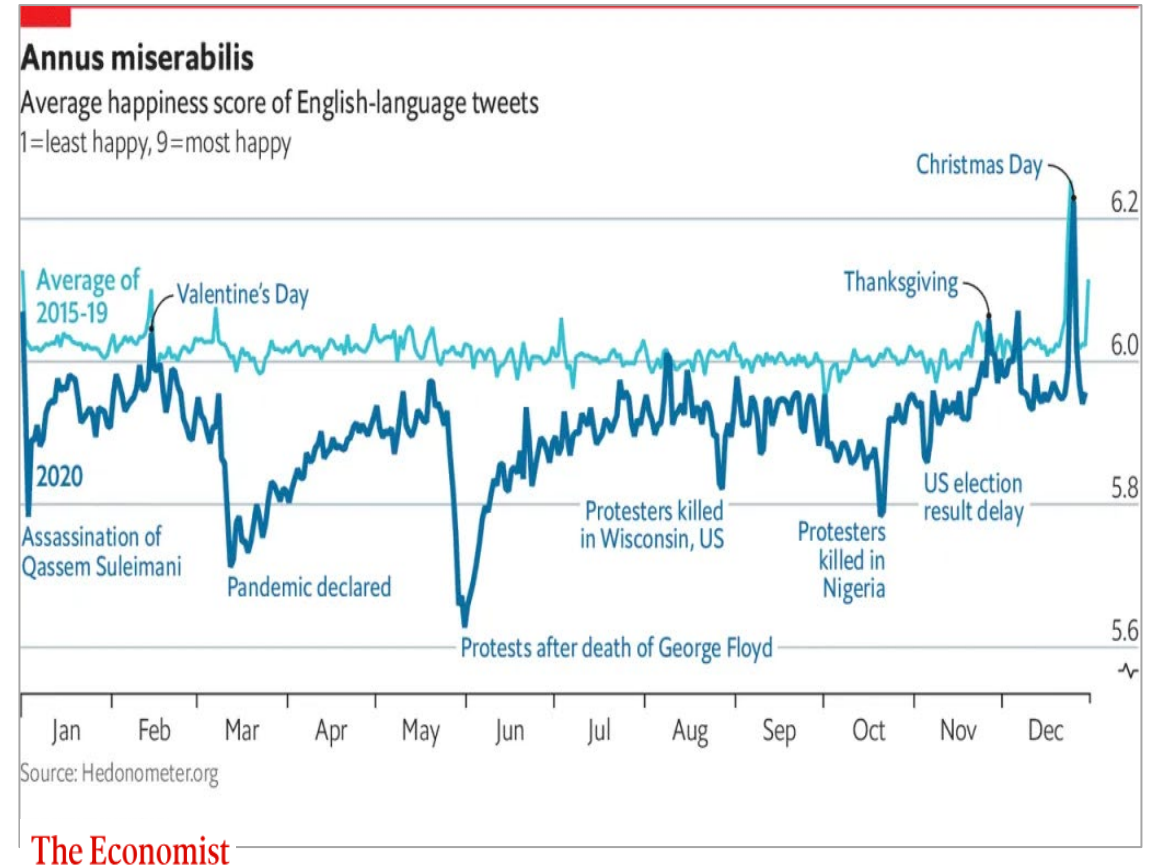
Publications. Paper 491 https://digitalcommons.usu.edu/itls_facpub/491/

2020

Data changes the **way we perceive and get informed** about the world, as well as understand complex phenomena.

Sentiment of 2020, the «miserable year».

2020 was a particularly intense year, but many things also happened between 2015 and 2019 that seem to have left us more undisturbed.



2024



Data Sources: wearables, lab results, patient metrics, and medical history.

Dynamic viz: personalised health insights (“Your resting heart rate is trending upward”), adaptive goal recommendations and progress visualisation.

Crowdsourcing

The volunteer contribution of people to an ad-hoc activity which typically consists in producing, collecting, sharing information on specific topics, is known as crowdsourcing.

J. Howe, **Crowdsourcing: Why the Power of the Crowd Is Driving the Future of Business**. Crown Publishing Group, 2008.

2005



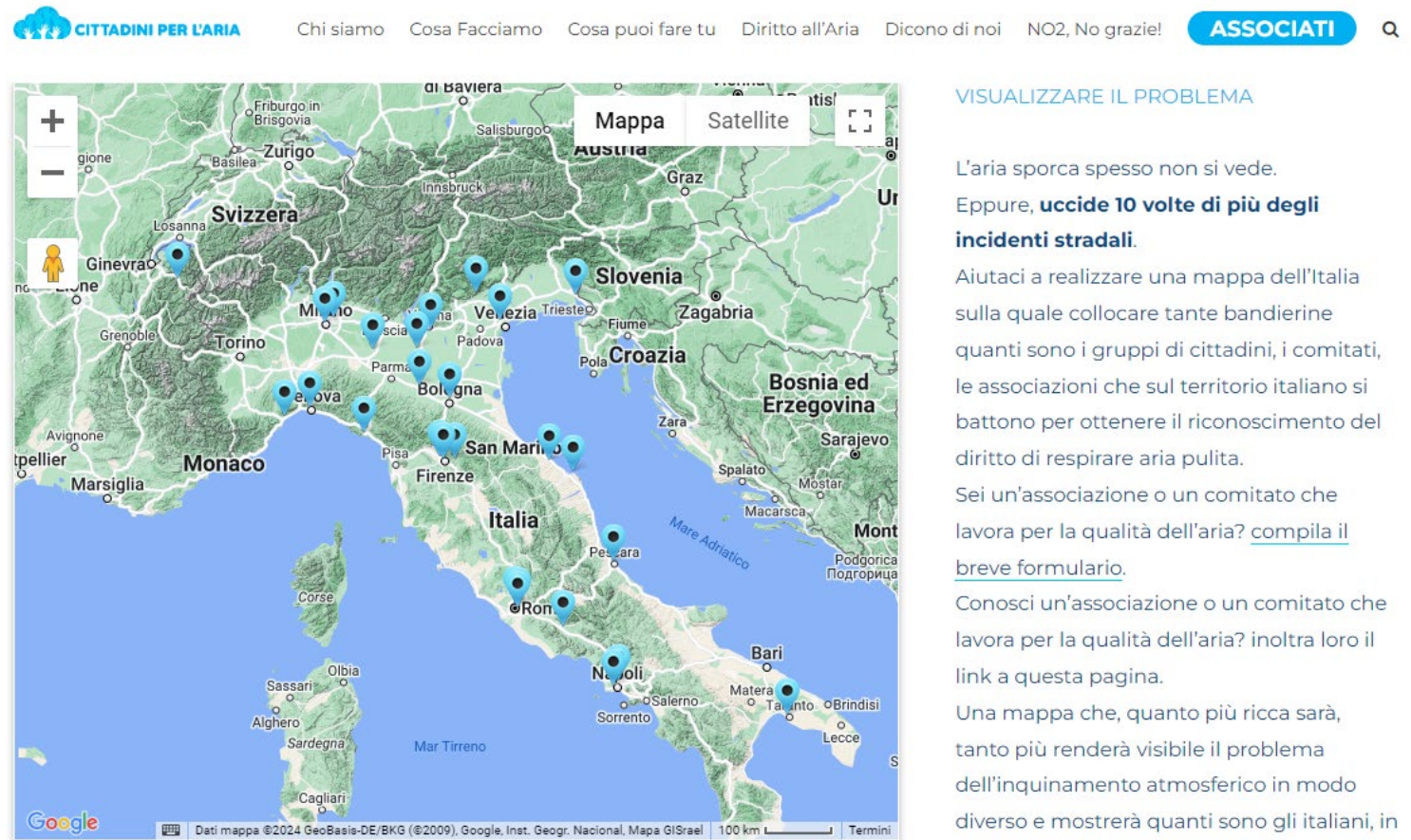
2013



Example > Cittadini per l'aria

Citizen-science projects, based on crowdsourced data on levels of nitrogen.

Studio Calibro designed and developed the map where citizens can read the nitrogen levels generated from diesel-powered cars and other sources.



Una RETE PER L'ARIA è una piattaforma che si propone di:

- attivare relazioni,

VISUALIZZARE IL PROBLEMA

L'aria sporca spesso non si vede.

Eppure, **uccide 10 volte di più degli incidenti stradali.**

Aiutaci a realizzare una mappa dell'Italia sulla quale collocare tante bandierine quanti sono i gruppi di cittadini, i comitati, le associazioni che sul territorio italiano si battono per ottenere il riconoscimento del diritto di respirare aria pulita. Sei un'associazione o un comitato che lavora per la qualità dell'aria? [compila il breve formulario.](#)

Conosci un'associazione o un comitato che lavora per la qualità dell'aria? inoltra loro il link a questa pagina.

Una mappa che, quanto più ricca sarà, tanto più renderà visibile il problema dell'inquinamento atmosferico in modo diverso e mostrerà quanti sono gli italiani, in ogni parte del paese, che considerano questo tema una priorità.

Grazie!

cittadiniperlaria.org/rete-italiana

Example > The Migrants' Files

*“We started The Migrants’ Files in mid-2013 after realizing that **there was no usable database** of people who died in their attempt to reach or stay in Europe. In the last three years, we collected data and used it to **shed light on government policy** in a domain that was driven by emotions rather than facts. Our data was reused by hundreds of news outlets, by artists, NGOs, activists and governments alike.”*

The Migrants' Files

The Migrants Files was a consortium of journalists from over 15 European countries. It was coordinated by **Journalism++**.



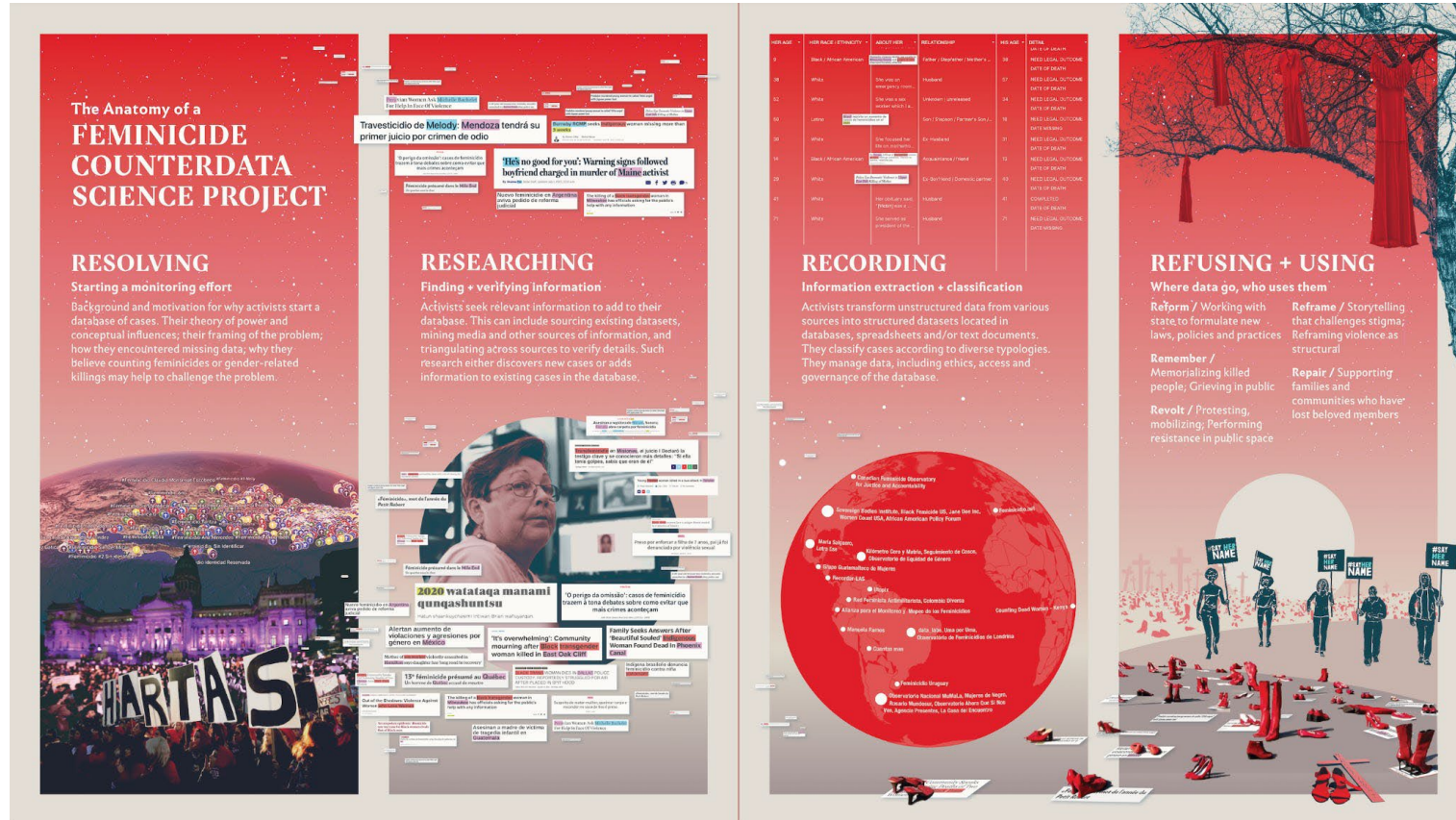
Winner, Data Journalism Awards 2014

Winner, European Press Prize 2015

An example of the so-called **counter-data** (datasets created and used to challenge, verify, or counterbalance existing data).

From data to the service

One team of data-journalists involved in project Uma-por-Uma, in the state of Pernambuco - Brazil, has combined the collection of data with a **proximity service for families of the victims**.



#UMAPORUMA

Onde encontrar ajuda

0800 281-8187 Ovidoria da Mulher do Estado de Pernambuco	180 Central de Atendimento à Mulher do governo federal	100 Central de Denúncia de Exploração Sexual e Tráfico de Mulheres do governo federal	190 - MULHER Polícia Militar de Pernambuco
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Centros de referências especializados

Centro Especializado de Atendimento à Mulher Clarice Lispector
Rua Bernardo Guimarães, nº 470 - Santo Amaro - Recife/PE - CEP: 50.050-440
Fone: (81) 3355-3009 / 0800-281-0107
Email: ordaricelispector@gmail.com

Centro Especializado de Atendimento à Mulher Márcia Dangremon
Rua Maria Ramos, nº 131 - Bairro Nova Olinda/PE - CEP: 53.020-010
Fone: (81) 3429-2007 / 0800-281-2008
Email: cmolinda@yahoo.com.br

Centro Especializado de Atendimento à Mulher Maristela Just
Rua Traversa São João, nº 64 - Prazeres - Massaranduba - Jaboatão dos Guararapes/PE - CEP: 54.310-091
Fone: (81) 3468-2485
Email: cm.maristelajust@gmail.com

Centro Especializado de Atendimento à Mulher Aqualtune
Rua Santa Tereza, nº 384, 1º andar - Centro - Paulista/PE - CEP: 53.000-000
Fone: (81) 3433-4447 / (81) 3437-0891
Email: bpinhoalves@hotmail.com

Centro Especializado de Atendimento à Mulher Maria Porcina Souto
Rua José Bezerra da Silva, nº 146
Centro - Cabo de Santo Agostinho/PE - CEP: 54.510-520
Fone: (81) 3518-1937
Email: cernap10@hotmail.com

Centro Especializado de Atendimento à Mulher
Avenida Dr. Pedro Correia de Araújo, 46
Centro - São Lourenço da Mata/PE - CEP: 54.735-210
Fone: (81) 98891-1488 / (81) 3535-3984
Email: tacitasilva@hotmail.com

Centro Especializado de Atendimento à Mulher
Rua Hilda da Costa Monteiro, nº 94 - Centro - Ipojuca/PE - CEP: 55.590-000
Email: cm.mulherdonaiairina.iipojuca@gmail.com

Centro Especializado de Atendimento à Mulher em Situação de Violência Doméstica e Sexista De Gravata
Rua João Pessoa, nº 170 - Centro - Gravata/PE - CEP: 55.642-000
Fone: (81) 3563-9003 Prefeitura
Email: secmulher@prefeitura.degravata.pe.gov.br

Centro Especializado de Atendimento à Mulher em Situação de Violência Doméstica e Sexista de Vertente do Lério
Rua Olímpia Batista de Santana, nº 22
Centro - Vertente do Lério/PE - CEP: 55.760-00
Fone: (81) 3634-7220 Prefeitura
Email: sec.mulhervertentelero@outlook.com

Centro Especializado de Atendimento à Mulher Maria Bonita
Rua Barão de Porto Seguro, nº 117
Maurício de Nassau - Canaúpe/PE - CEP: 55.004-350
Fone: (81) 3721-0499
Email: karimnyoliveira@canaupe.pe.gov.br

Centro Especializado de Atendimento à Mulher Josefa Mendes de Barros
Rua José Moraes da Silva, nº 230
Centro - Santa Cruz do Capibaribe/PE - CEP: 55.190-000
Fone: (81) 3731-1826 Prefeitura
Email: coordenadoriadomesticasecrista@gmail.com

Centro Especializado de Atendimento à Mulher em Situação de Violência Doméstica e Sexista Beatriz Lima e Silva
Rua Frei Caneca, nº 344 - Heliópolis - Garanhuns/PE - CEP: 55.642-915
Email: cramgaranhuns@gmail.com

Centro Especializado de Atendimento à Mulher em Situação de Violência Doméstica e Sexista De Agrestina
Rua Coronel Manoel Alves, nº 110
Centro - Agrestina/PE - CEP: 55.495-000
Fone: (81) 3744-1103 Prefeitura
Email: secretariamulheragrestina@gmail.com

Data is a need.

In critical conditions, regardless of the type of crisis and emergency, people need to exchange information, both receiving and providing it.

They use all the known channels (Social media, Maps-based services, IM)

#Chennai Rains/The W...

Mark off areas, streets and small roads that are water-logged and impossible to cross by any means of private

220,944 views

SHARE

Water-Logged Roads

- Velachery Bypass Road
- Srinivasa Road - T Nagar
- Chettnad Health City Bus Stop
- Keelkattalai Junction
- ... 36 more

Road Blocks-Slow Traffic

- Siddo Nagar
- Buzuliah Road - T Nagar
- Madhya Kallesh Temple
- Avvai Shanmugam Salai

Chennai Flooding

FACEBOOK SAFETY CHECK

Quickly find and connect with friends in the area. Mark them safe if you know that they're OK.

Are you in the affected area? Yes, let my friends know

FOR HELP IN #CHENNAIRAINS: HERE ARE THE HASHTAGS YOU NEED TO KNOW

#CHENNAIRAINSHelp
If you know of someone, or you yourself require help or want to offer help, please use this hashtag, tagging @TwitterIndia so we can help RT

#CHENNAIVOLUNTEER
If you have supplies or can provide volunteer support, please use this hashtag

#CHENNAIRESCUE
If you are stranded and in desperate need of rescue, please use this hashtag & tag @TwitterIndia so we can RT

Crowdsourced list of places and people offering shelter

Area in Chennai where shelter is available	Accommodation available for (No. of people)	Twitter Handle/Facebook Profile	Mobile Number	Original Source Link for information
Tambaram, Perungudi, Vandalur and Urapakkam		@saurabh		Good help at Tambaram, Perungudi
100 Feet Road	400	@saurabh	9190742958	SELF
Adyar	5	@praveenprakash	900061122	SELF
Adyar	5	praveenprakash	900061122	SELF

Shelter and Food

Offering shelter for 4 people
Mumbai

Pushparaj
@pushparaj15

I live near Thiruvannamiyur RTO. I can accomodate two ppl in my home. Contact : 9994476633 #ChennaiRainsHelp #ChennaiFloods

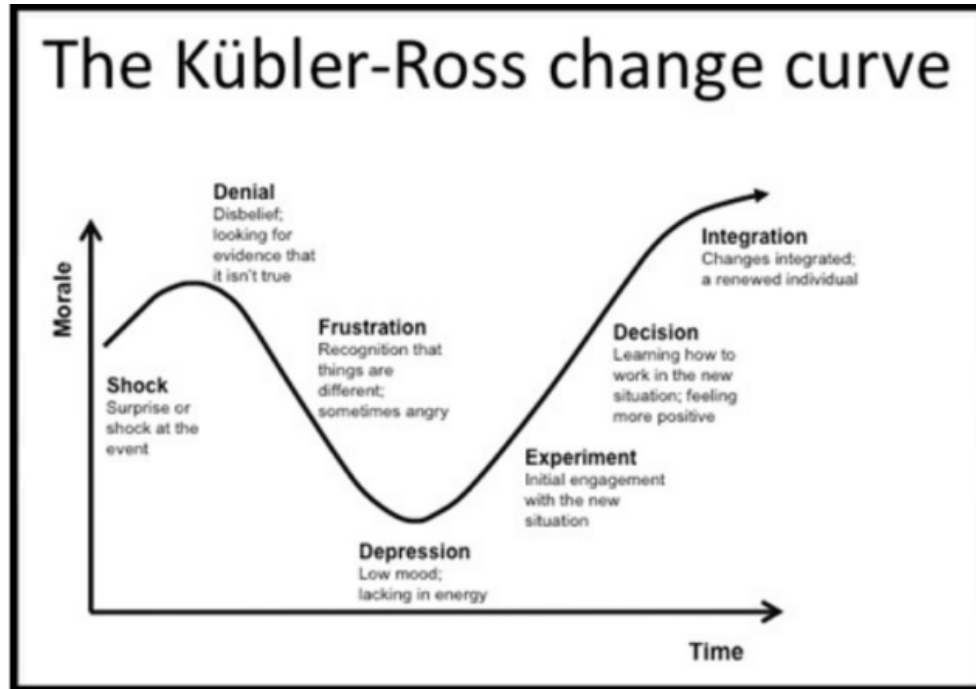
RETWEETS 210 LIKES 35

11:04 PM - 1 Dec 2015

Chennai flood, 2015

A. Olteanu, S. Vieweg, and C. Castillo, **What to expect when the unexpected happens: Social media communications across crises**, in Conference on Computer Supported Cooperative Work; Social Computing. ACM, 2015, pp. 994–1009.

Human factors + context = diverse (data) experiences.



<https://www.ekrfoundation.org/5-stages-of-grief/5-stages-grief/>

People (we) don't interact with data in a vacuum.

The person's **emotional state**, individual traits (such as prior knowledge, motivations, and biases), plans, data literacy, and the context in which they are some conditions that determine how data are **received, comprehended, and put into action.**

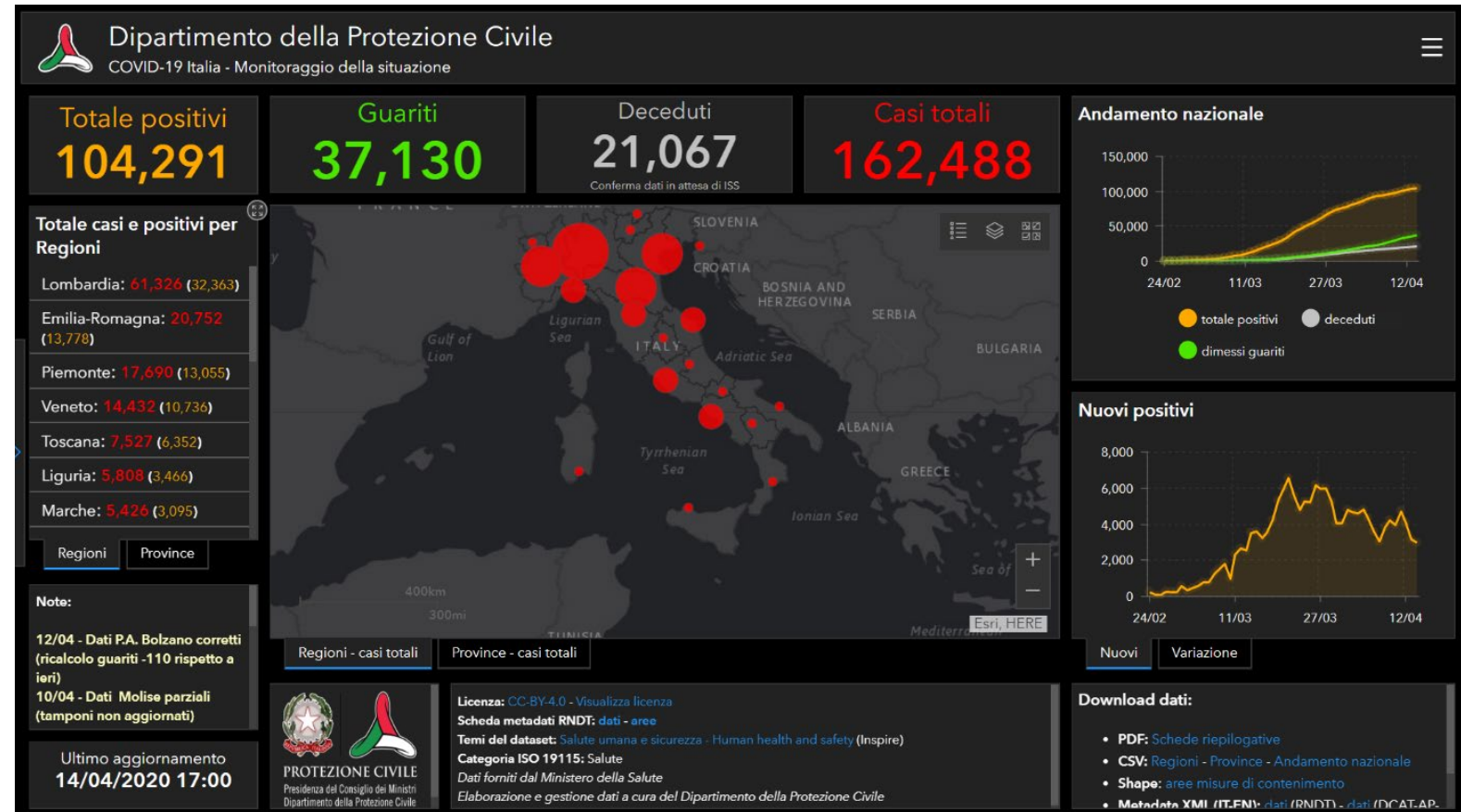
Good design can improve it.

Tong, J. (2023). **From content to context: A qualitative case study of factors influencing audience perception of the trustworthiness of COVID-19 data visualisations** in UK newspaper coverage. *Journalism*, 25(7), 1481-1499. <https://doi.org/10.1177/14648849231190725>

Data shapes phenomena.

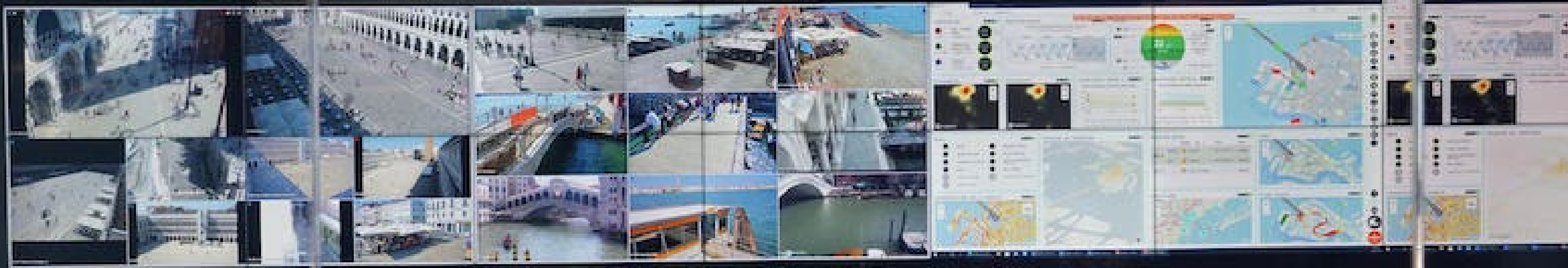
Web-accessible dashboards allow different types of actors to **view data and monitor complex phenomena.**

- Monitoring complex events/systems (total visibility)
- Trend identification
- Measure of efficiency
- Decision support
- Automated reports
- Fast identification of abnormal data and correlations
- Alert



data.europa.eu

Dashboard for professionals (into the wild).



Dashboard for professionals (into the wild).



Perception
(event detection, search)

Comprehension
(comparison, aggregation
of info)

Projection
(evolution of the scenario)

**Decision and
coordination**
(intervention
execution and
monitoring)

Situation awareness

the perception of the elements in the environment within a **volume of time and space**,
the comprehension of their meaning, and the projection of their status in the near future.

Quasi-professionals dashboards.



Plane control room

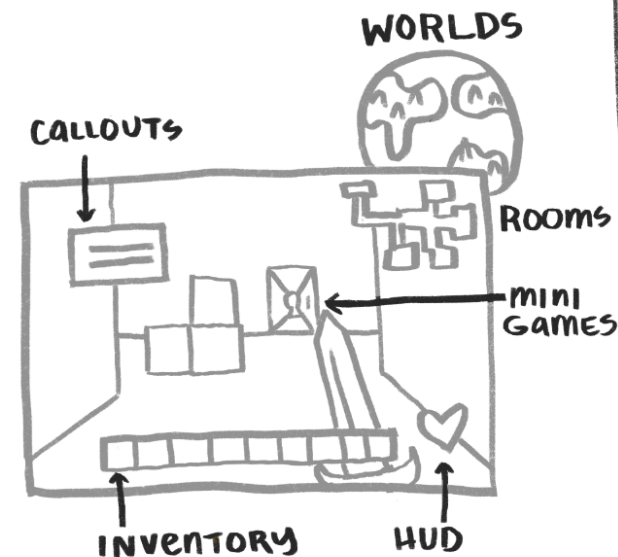


Airspace explorer (separation data)

Dashboard for playing/training.



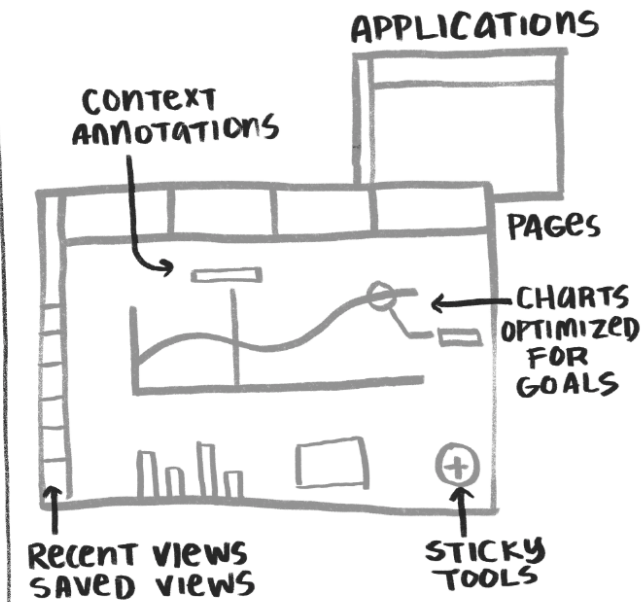
game design



CHARACTERS



DASHBOARDS



USER TYPES



Susie Lu. [Game Design Deep Dive: Situational awareness and player frustration in GRIP](#)

Dashboards for the public.

Sostenibilità Osservatorio

f

in

Temi Caldi

Criptovalute Spread BTP-Bund FTSE-MIB Petrolio

24+ ABBONATI

Publicità

Osservatorio ESG

Indagine sulla situazione della Sostenibilità (Ambiente, Sociale e Governance) nelle piccole e medie società italiane quotate a Piazza Affari e sul mercato Aim.

Informazioni generali

n° ESG/Aziende 2023

136/358

n° ESG/Aziende 2022

146/351

n° ESG/Aziende 2021

117/332

Regione sede legale (top 5)

Lombardia

Lazio

Emilia-Romagna

Veneto

Toscana

Altre regioni

Estero

46

19

13

11

8

19

1

2023

2022

2021

Settore economico (top 5)

Industrials

Consumer Cyclicals

Technology

Financials

Telecommunications

Altri settori

27

23

10

16

19

22

2023

2022

2021

Temi Caldi

Criptovalute Spread BTP-Bund FTSE-MIB Petrolio

24+ ABBONATI

Ambiente

Sociale

Governance

Controversie

Ambiente

In questa sezione si vanno ad analizzare le policy ambientali, il grado di dettaglio e la struttura messa in atto dall'azienda per realizzarle, evidenziando inoltre gli obiettivi e l'analisi degli impatti generati.

Policy ambientale

100%

90%

80%

70%

60%

50%

40%

30%

20%

10%

0%

98%

91%

99%

2023

2022

2021

2023

2022

2021

SI

No

131

5

133

13

115

1

Target, obiettivi e scadenze ambientali

100%

90%

80%

70%

60%

50%

40%

30%

20%

10%

0%

39%

35%

33%

61%

65%

67%

2023

2022

2021

2023

2022

2021

SI

No

80

51

86

46

78

38

Impegno utilizzo di energie rinnovabili

21%

79%

2023

2022

2021

Impegno verso l'efficienza energetica

12%

88%

2023

2022

2021

Impegno a ridurre le emissioni (Green House Gas)

24%

76%

2023

2022

2021

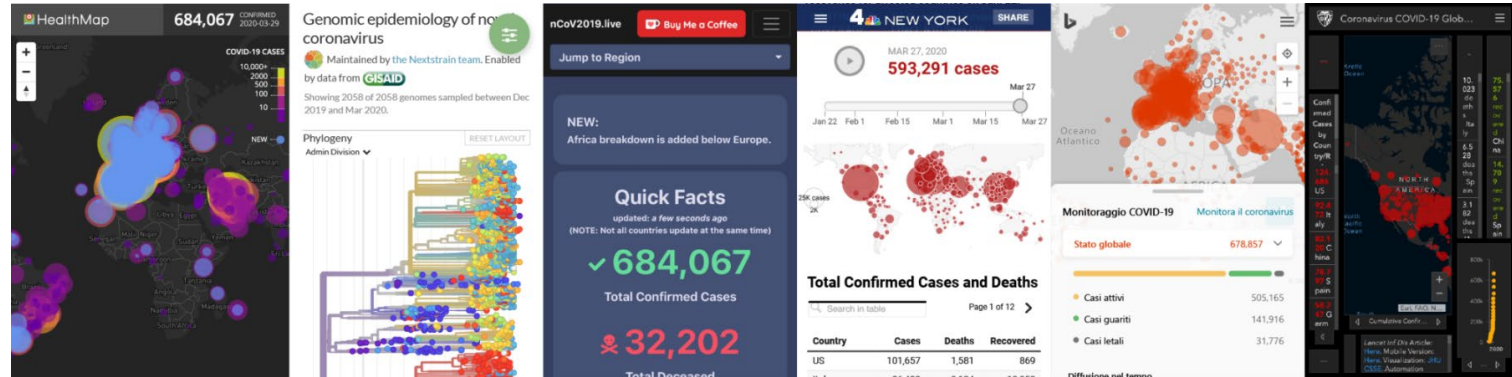
Questionari

Cerca azienda per nome o codice ISIN...

AZIENDA	REGIONE	SETTORE	QUESTIONARI
ABC Company IT0005446294	Lombardia	Financials	2023 2022 2021
Abitare In IT0005445280	Lombardia	Consumer Cyclicals	2023 2022 2021
Acea IT0001207098	Lazio	Utilities	2023 2022 2021
Aeffe IT0001384590	Emilia-Romagna	Consumer Cyclicals	2023 2022 2021
ALA IT0005446700	Campania	Industrials	2023 2022 2021
Alfonso IT0005446039	Campania	Technology	2023 2022 2021
Alkeny IT0005314635	Lombardia	Consumer Cyclicals	2023 2022 2021
Alcore IT0005461303	Lombardia	Industrials	2023 2022 2021
Almawave IT0005434615	Lazio	Technology	2023 2022 2021
Altea Green Power IT0005472730	Piemonte	Energy	2023 2022 2021

Dashboard for the public.

Designed by Matteo Moretti,
Created for the community, to face
the mess of data and data sources
available during the pandemic. The
dashboard plot the data from the
Civil Protection on the virus spread
to make it more **readable**,
comparable on different
dimensions and **interactive**.

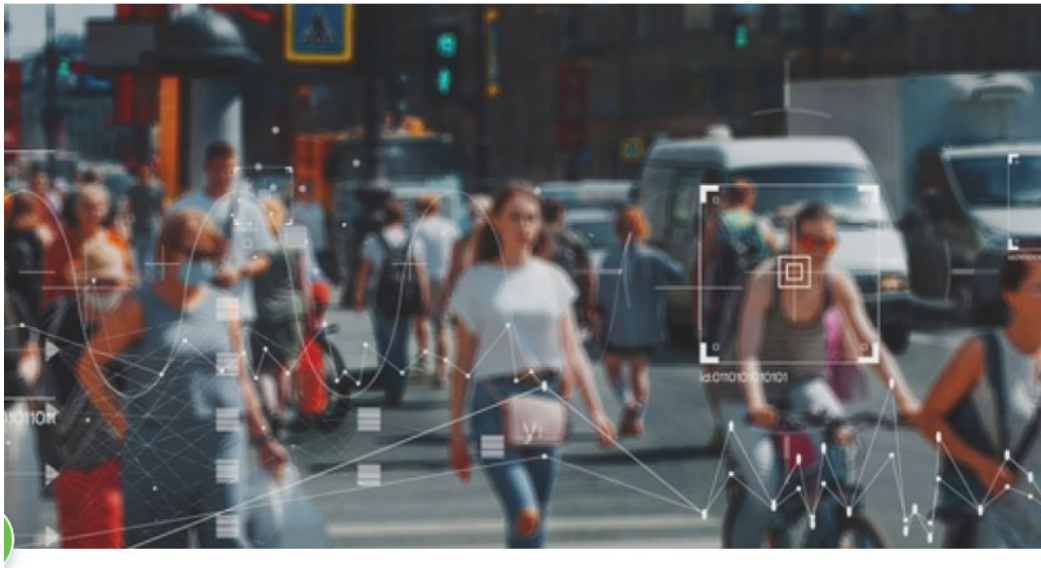


Data shape decisions.

NYC's Data-Driven Future: 46 Algorithms and Counting

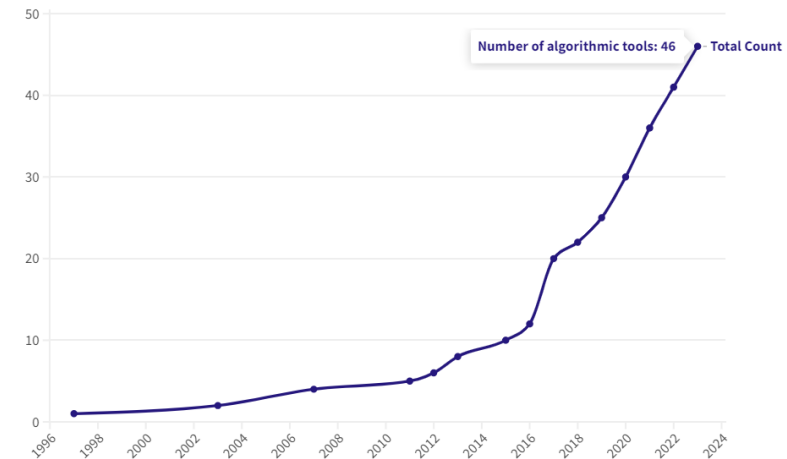
New York City, which tracks the number of algorithms its departments use, has seen adoption of those tools skyrocket in recent years. Officials see them as a way to modernize government and enhance efficiency.

July 31, 2024 • Nikki Davidson



Evolution of Algorithmic Tool Usage in New York City

New York City's use of algorithmic tools has grown steadily since 1997, skyrocketing since 2016.



Source: NYC Compliance Reporting of Algorithmic Tools 2020-2023 • Created July 30, 2024. Visualization by Nikki Davidson.

* A Flourish chart

Data shape decisions.

NYC's Data-Driven Future: 46 Algorithms and Counting

DEPARTMENTS:

- **Child welfare:** to predict future risks, help the Administration **prioritize interventions** and **allocate resources** more effectively.
- **Education:** to **assess educator performance**, informing personnel decisions.
- **Public health:** to **monitor social media** and **311 complaints** for signs of foodborne **illness outbreaks**, while other tools analyze **wastewater** and **genetic data** to **predict disease risks** and **optimize public health programs**.
- **Emergency response:** to **match patients** with the nearest appropriate **hospitals**, optimize **ambulance dispatching**, predict **peak times for emergency calls** and prioritize fire inspections based on risk assessment.
- **Environmental protection:** to identify **idling vehicles**, monitor air quality.

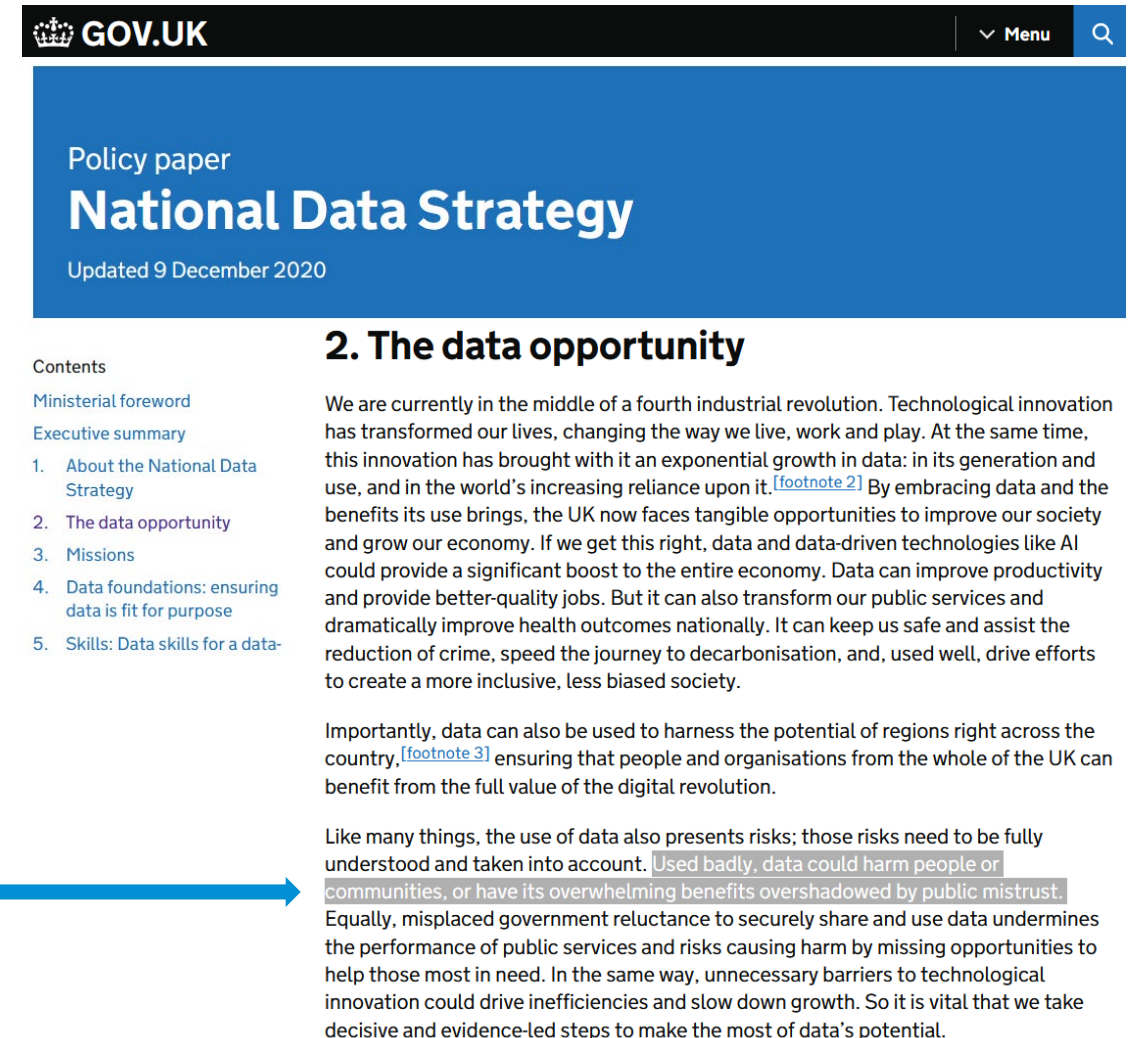


Source: [NYC Compliance Reporting of Algorithmic Tools 2023](#) • Created July 30, 2024. Visualization by Nikki Davidson.

* A Flourish bubble chart

Data is a human right.

The UK in 2020 formalized a National Data Strategy, which says that **citizens must have the right to "know" how to use data.**



The screenshot shows the official UK Government page for the National Data Strategy. The header includes the GOV.UK logo and a navigation menu. The main title is "National Data Strategy" with a subtitle "Policy paper" and a date "Updated 9 December 2020". A table of contents lists sections from the Ministerial foreword to Skills. The main content area is titled "2. The data opportunity" and contains two paragraphs. The first paragraph discusses the fourth industrial revolution and the benefits of data. The second paragraph discusses the potential of data for regions and the need to ensure it is used safely. A blue arrow points from the text "Data is a human right." to the second paragraph.

GOV.UK

Policy paper

National Data Strategy

Updated 9 December 2020

Contents

- Ministerial foreword
- Executive summary
- 1. About the National Data Strategy
- 2. The data opportunity
- 3. Missions
- 4. Data foundations: ensuring data is fit for purpose
- 5. Skills: Data skills for a data-

2. The data opportunity

We are currently in the middle of a fourth industrial revolution. Technological innovation has transformed our lives, changing the way we live, work and play. At the same time, this innovation has brought with it an exponential growth in data: in its generation and use, and in the world's increasing reliance upon it.^[footnote 2] By embracing data and the benefits its use brings, the UK now faces tangible opportunities to improve our society and grow our economy. If we get this right, data and data-driven technologies like AI could provide a significant boost to the entire economy. Data can improve productivity and provide better-quality jobs. But it can also transform our public services and dramatically improve health outcomes nationally. It can keep us safe and assist the reduction of crime, speed the journey to decarbonisation, and, used well, drive efforts to create a more inclusive, less biased society.

Importantly, data can also be used to harness the potential of regions right across the country,^[footnote 3] ensuring that people and organisations from the whole of the UK can benefit from the full value of the digital revolution.

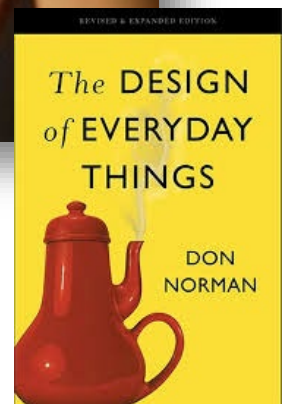
Like many things, the use of data also presents risks; those risks need to be fully understood and taken into account. **Used badly, data could harm people or communities, or have its overwhelming benefits overshadowed by public mistrust.** Equally, misplaced government reluctance to securely share and use data undermines the performance of public services and risks causing harm by missing opportunities to help those most in need. In the same way, unnecessary barriers to technological innovation could drive inefficiencies and slow down growth. So it is vital that we take decisive and evidence-led steps to make the most of data's potential.

A collection of various 3D isometric charts and graphs, including bar charts, pie charts, donut charts, and line graphs, rendered in vibrant colors like red, orange, yellow, green, and blue. The charts are arranged on a dark, textured background. Some charts include labels like '1.', '2.', '3.', '4.', '01', '02', '03', '04', 'A', 'B', 'C', 'D', and '100%'. The styles are modern and colorful, with some charts featuring multiple series or segments.

*“Artifacts pervade our lives, our every activity. Technology, potentially, makes our daily life more comfortable and pleasant... But at the same time, **it adds complexity and complication**”.*

TECHNOLOGY PARADOX

*Complexity and difficulties are inevitable when increasing the number of features. **A good design can minimise it**”.*



From a human-centred perspective, data is...

Phenomena that occur and that begin to exist as data **if and when** someone decides to **observe, count and classify** them.

Data is a **construction** and reflects the relevance given to what it is decided to be measured.

Data must answer questions and serve **goals**.

Data is something that we use to reduce uncertainty and make better predictions and decisions.

The data exists in relation to the subjects that **produce, collect, use it**, whether it is a single individual or a research community.



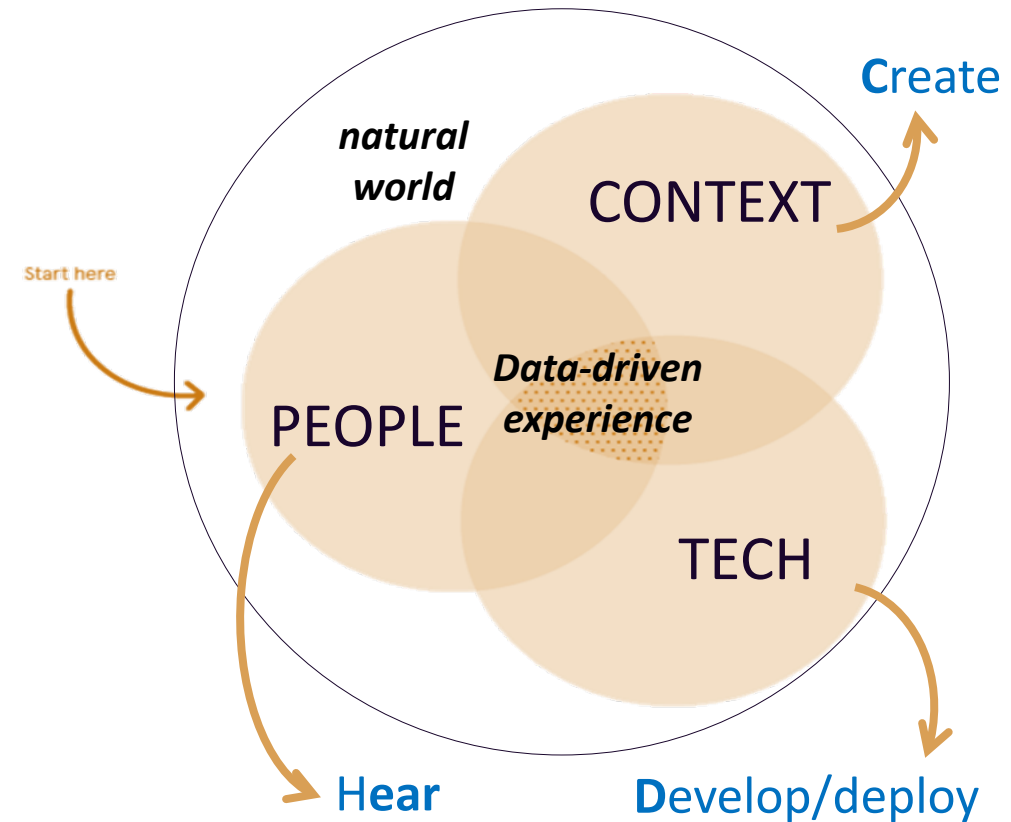
The Human Centred Design approach (HCD).

Design approach defined by the seminal work of Donald Norman, psychologist and Computer Engineer @MIT.

He reinterpreted the Human-computer Interaction (HCI) in a psychological key, widening the **focus** from technology on the **people who use it**.

In this perspective, **EVERY SYSTEM INCLUDES PEOPLE** interacting with it in a specific context.
Here, data and information work as **SOCIAL CONSTRUCTS** that human beings generate, use, exchange, and enrich them.

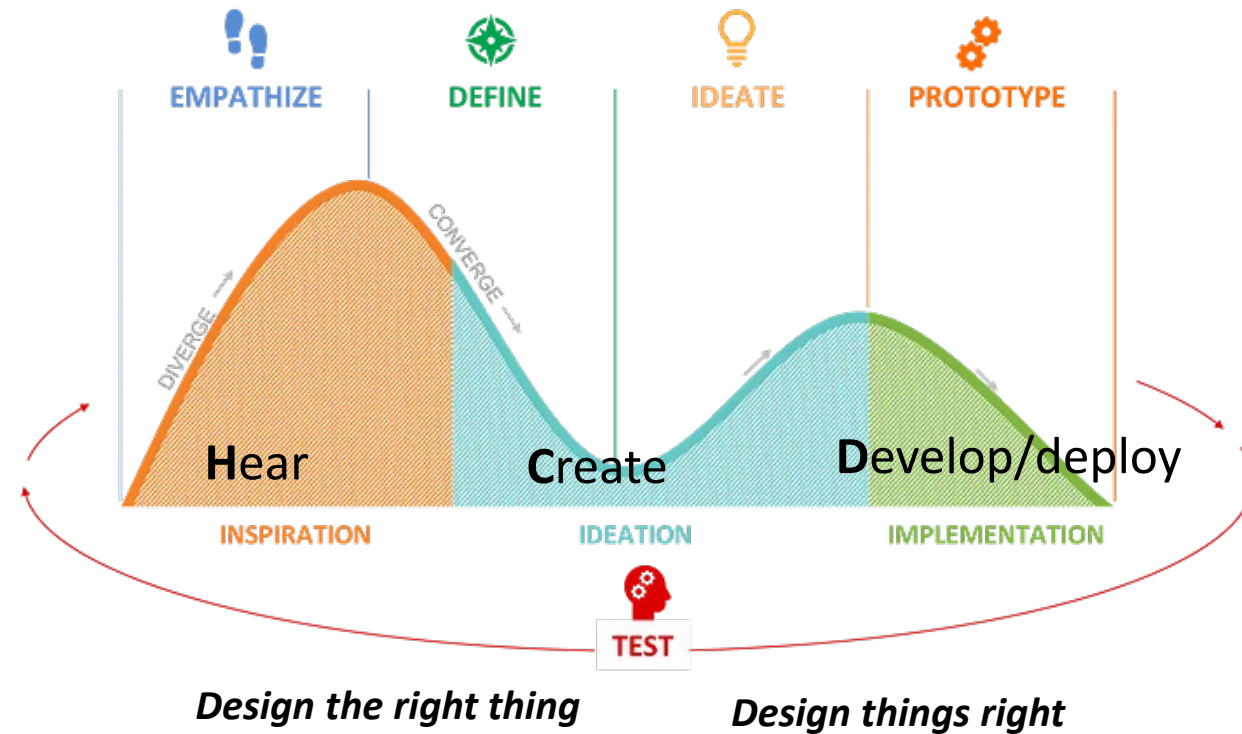
This approach is the common background of many methodologies worldwide applied in the fields of software development (**Agile, Lean, Scrum**), design (**Design Thinking, UXD, Sprint**), education, social innovation, and other areas.
It is also formalised in several ISO Standards.



Norman, D.A. & Draper, S.W. **User centered system design. New perspectives on human-computer interaction**, 1996

Design thinking. In practice.

- **Problem setting** before. Problem-solving after.
To reply to actual needs design before developing.
- **Cooperate** with stakeholders:
incorporate needs and perspectives of **direct and indirect beneficiaries**.
- **Fail** early and often. Test, learn from errors and design again.
- **Show, don't tell**: prototyping is more than testing, it's space to make **mistakes**, learn, progressively improve, and take better decisions.
- **Test and Iterate**



HCD | Steps and tools

*The macro-process scales up the core process:
in every step, you will learn, ideate and test.*

H HEAR: Study, Visit, Listen, Observe, Ask

- Define the problem
- Map the context
- Know user characteristics and needs

C CREATE, compare, draw, sketch, prototype

- Design big vision and details before starting to develop
- Design the interaction
- Test alternatives

D DEVELOP, Test, Refine, Re-test, Describe, Evolve

- Test the development
- Test with users
- Also design the support

Tools to:
Know, Design, Share, Refine

DATA COLLECTION

+

DATA DISPLAY

H

DATA SET
N/ETNOGRAPHY
CROWDSOURCING
INTERVIEWS

STAKEHOLDERS MAP
PERSONAS
EMPATHY MAP

C

VALUE PROPOSITION
TECHNICAL REQUIREMENTS

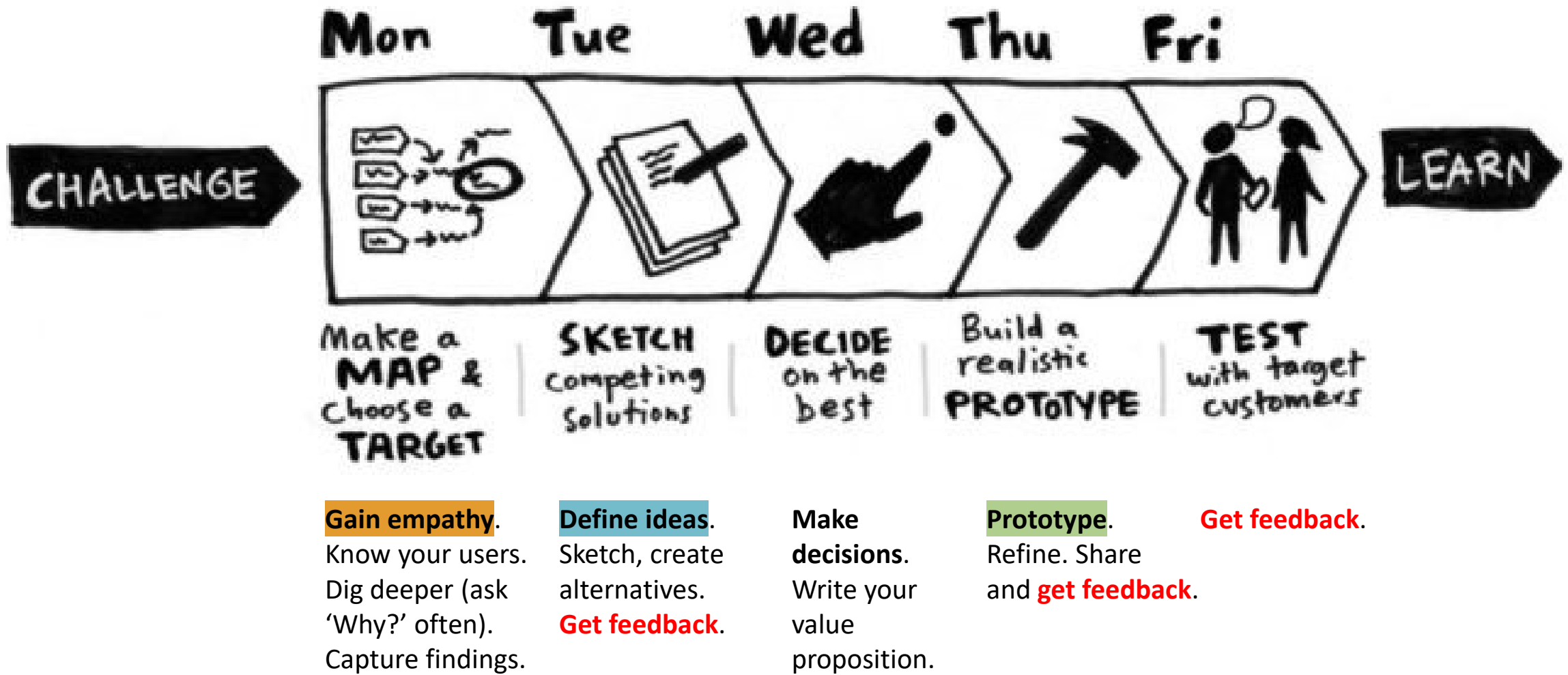
JOURNEY MAP
INFO-ARCHITECTURE
SKETCHES, FLOW CHARTS

D

HEURISTIC EVALUATION
EXPERT ASSESSMENT
COGNITIVE WALKTHROUGH
USER FEEDBACK

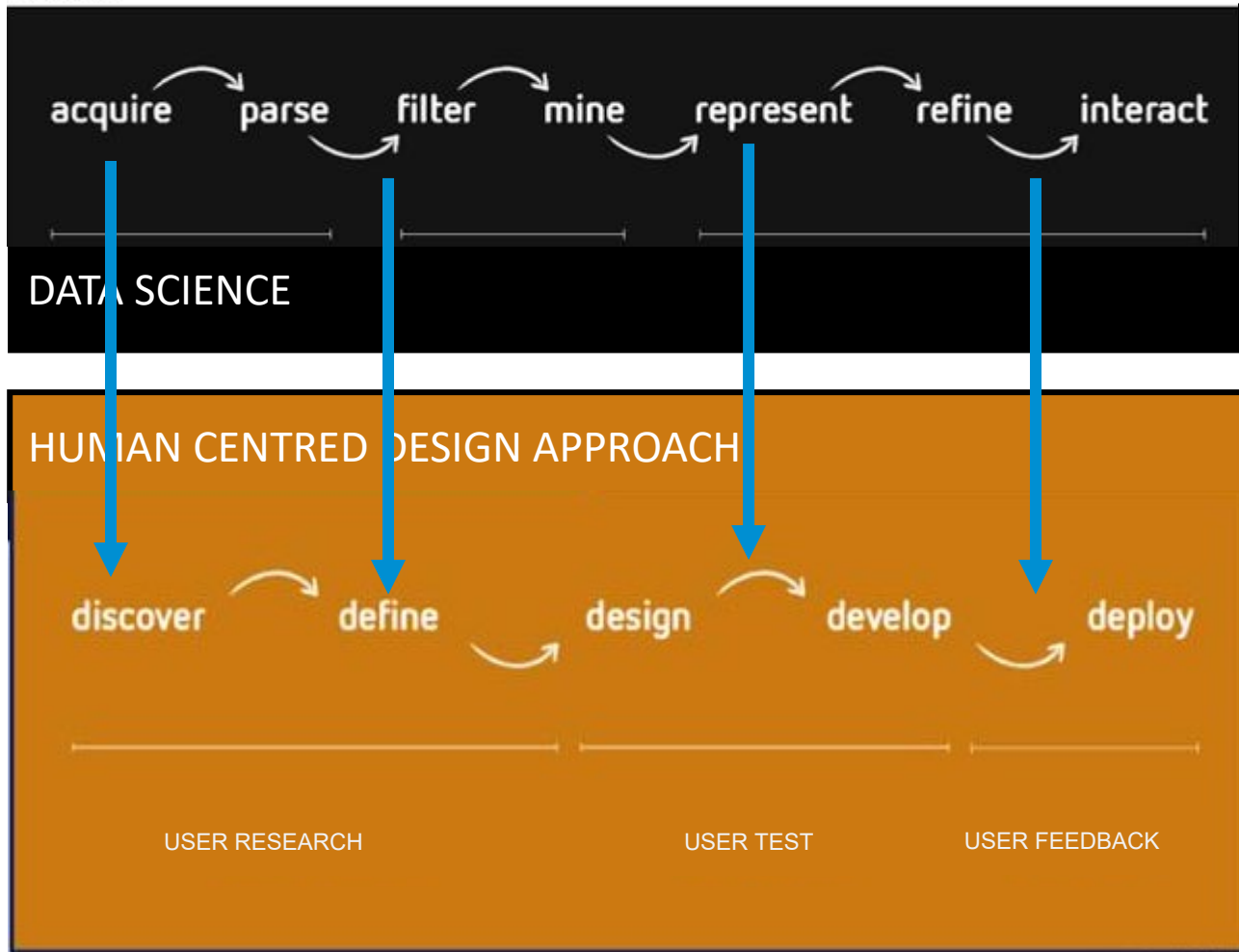
PoC
WORKING PROTOTYPES
USER GUIDE

Sprint your design.



Data science || Human Centred Design (HCD)

Fry (2004).



IDEO (2015).



Contacts

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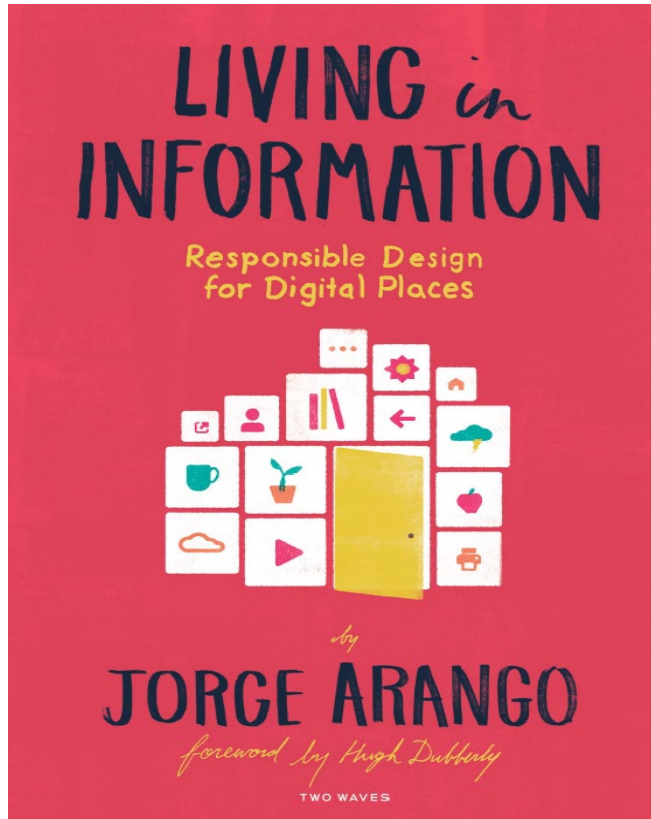
antonella.frisiello@polito.it



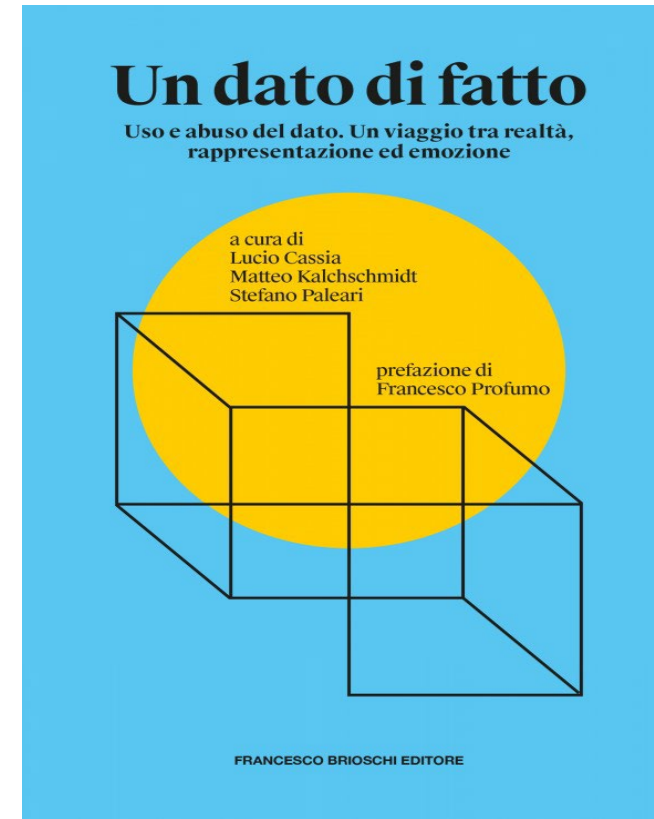
**Politecnico
di Torino**

Additional resources

To deep dive the data culture.



Jorge Arango. [Living in Information. Responsible Design for Digital Places.](#) 2018



Lucio Cassia et al.,
Un dato di fatto.

What does it mean to count?

The German anthropologist Karl Menninger defines counting as
“**assigning number words to things.**”

What do we do when we actually count?

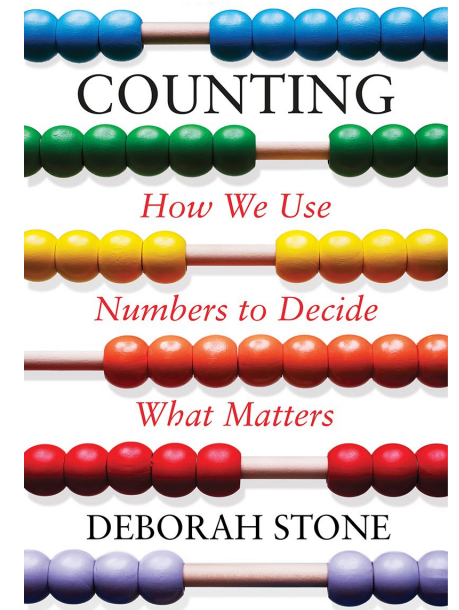
Counting meets power when force things into categories by ignoring their differences and ultimately, their uniqueness

Quantitative thinking, like qualitative, is highly cultural.

The hard part is learning to see the likenesses your culture deems important and to ignore those it deems incidental.

Numbers get their meaning partly from verbal and visual cues, the kinds of devices Darryl Huff unmasked in *How to Lie With Statistics* and Edward Tufte does in his books on **infographics**.

“Numbers are fragments of our imagination, *fictions* really,
no more true than poems or drawings.
In this sense, *all* statistics are lies.”



Deborah Stone, [The ethics of counting](#), 2017.

Also in: **Counting**, how we use numbers to decide what matters 2020.